Connecting the Dots in AMR Surveillance

How bacteria climb the resistance ladder, and how to forecast the next rung

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Abstract

Antibiotic resistance is often studied one species–antibiotic pair at a time, even though resistance mechanisms can affect multiple drugs and distinct determinants can accumulate or be transmitted together [1, 2]. This pairwise view captures local resistance levels but overlooks the broader organization of resistance across species and antibiotics. Building on evidence that resistance capabilities can accumulate along partially constrained trajectories, we ask whether surveillance data reveal a system-wide resistance structure and whether this structure can help anticipate future resistance events. We use data from the Pfizer ATLAS programme distributed through the Vivli AMR Register, comprising 394 species, 43 antibiotics, and observations from 2004 to 2024 [3]. Rather than analysing absolute resistance alone, we measure whether each species is more resistant to a drug than expected from both the species' overall resistance level and the drug's overall failure rate. These comparative-resistance links define a bipartite network connecting species and antibiotics.

We find that the network is strongly nested within coherent organism–drug groups: species linked to few antibiotics are generally linked to a subset of those reached by species with broader resistance patterns. This creates a ladder-like structure, in which species occupy progressively higher rungs as they become linked to additional, less widely defeated drugs. We then apply the Fitness–Complexity framework, originally developed to rank countries and products in similarly nested economic networks, where countries accumulate productive capabilities and progressively access more complex products [4]. Here, the method ranks species according to the breadth and difficulty of the drugs they defeat, and antibiotics according to the resistance capabilities associated with defeating them. Although clinical labels are not used in the construction, antibiotic complexity increases across the WHO Access, Watch, and Reserve tiers, while ESKAPEE taxa occupy the upper rungs of their local ladders [5, 6].

Finally, we connect antibiotics defeated by the same species to construct an empirical similarity network. A parameter-free exposure score derived from this network achieves an AUC of 0.71 for predicting new comparative-resistance links and remains informative over horizons of up to five years. We present the framework as an interactive atlas (`https://giordano-demarzo.github.io/resistome-atlas/`) that provides, for each pathogen, a system-wide view of its position on the resistance ladder and the drugs it is closest to defeating. Source code and reproducibility materials are available on GitHub [7].

Objectives

We pursued three linked objectives: (*i*) turn routine counts into a pathogen–antibiotic network that isolates pair-specific comparative resistance from the general resistance level of a species and the general vulnerability of a drug; (*ii*) test whether comparative-resistance profiles are nested, as partially ordered accumulation of mechanisms predicts [8, 9], and use Fitness–Complexity to place every species and drug on the resulting ladder; and (*iii*) test prospectively whether a species' position relative to empirically similar drugs anticipates which drug it defeats next.

Methods

From surveillance counts to a pathogen–antibiotic network

The input contains annual species–antibiotic counts from the Pfizer ATLAS surveillance program distributed through the Vivli AMR Register, spanning 394 species, 43 antibiotics and 2004–2024 [3]. The resistance fraction for each pair was defined as $r_{pa} = n^R_{pa}/N_{pa}$. A pair was considered sufficiently observed when at least ($n_{\min} = 30$) isolates had been tested. Counts were pooled over each analysis window before this threshold was applied. We denote the set of sufficiently observed pairs by $\mathcal{O}$. The workflow was repeated at $n_{\min} = 10$ and 50, all else unchanged. Absolute resistance may reflect either broadly resistant species, broadly ineffective antibiotics, or an unusually resistant species and antibiotic pair. We focus on the third component after accounting for both margins.

To quantify this comparative signal, we first define the two margins of the observed resistance matrix. For sufficiently observed pairs, define the species margin $R_p =$, antibiotic margin A_a, and total resistance T as $R_p = \sum_{a:(p,a)\in\mathcal{O}} r_{pa}$, $A_a = \sum_{p:(p,a)\in\mathcal{O}} r_{pa}$ and $T = \sum_{(p,a)\in\mathcal{O}} r_{pa}$. We thus can define the revealed comparative advantage of species p against antibiotic a as

$$\mathrm{RCA}_{pa} = \frac{r_{pa}}{R_p A_a / T} = \frac{r_{pa} T}{R_p A_a}\,.$$

Thus, $\mathrm{RCA}_{pa} = 2$ indicates that the observed resistance fraction is twice the value expected from the species and antibiotic margins. Adapted from economic complexity [10], this measure is invariant to proportional increases across the entire matrix and therefore isolates comparative rather than overall resistance.

We define a comparative resistance link when $\mathrm{RCA}_{pa} \geq 1$. The binary matrix $M_{pa} = \mathbf{1}\{\mathrm{RCA}_{pa} \geq 1\}$ defines a bipartite comparative-resistance network, with species on one side and antibiotics on the other. Everything that follows is computed from this object. For insufficiently observed pairs, $\mathrm{RCA}_{pa} < 1$is undefined and $M_{pa} = 0$; thus, zero does not imply confirmed susceptibility, and RCA should not be interpreted as a clinical breakpoint.

Nested profiles: the resistance ladder

If resistance mechanisms accumulate along partially constrained trajectories, the binary comparative-resistance matrix M should have a characteristic shape [8, 9]. If resistance capabilities accumulate along partially constrained trajectories, narrower comparative resistance profiles should be contained within broader ones, producing an approximately triangular matrix. This structure is known as *nestedness*, common in ecological bipartite networks [11, 12]. In our setting, nestedness provides the structural signature of the resistance ladder: comparative resistance profiles form successive rungs of a broadly ordered accumulation process rather than arbitrary sets of antibiotics. Because each analysis window is observational, however, nestedness is only consistent with an accumulation process and does not establish a temporal sequence or identify the mechanisms responsible for it.

Nestedness is biologically interpretable only among species exposed to broadly comparable antibiotic spectra. Intrinsic barriers, resistance mechanisms, and testing practices differ across taxa: for example, the Gram negative outer membrane limits the entry of many large antibiotics, while susceptibility interpretation rules also vary across organisms [13, 14]. We therefore did not assume that Gram positive and negative organisms occupied a single ladder, but partitioned (M) by spectral co clustering into five matched species and drug groups [15]. The partition used no taxonomy, drug class, AWaRe, or ESKAPEE labels.

For species i, j with degrees k_i, k_j, let $O_{ij} = \sum_a M_{ia} M_{ja}$ be the number of drugs both defeat, with degree-based expectation $\langle O_{ij} \rangle = k_i k_j / A$ over A drug columns. The statistic compares the overlap of a narrower profile with a broader one against this expectation, and analogously for drugs [16, 17]:

$$\mathcal{I}_R = \sum_{\substack{i,j \\ k_i > k_j,\, b_i = b_j}} \frac{O_{ij} - k_i k_j / A}{k_j (C_i - 1)} \quad , \qquad \mathcal{I}_C = \sum_{\substack{a,c \\ k_a > k_c,\, b_a = b_c}} \frac{O_{ac} - k_a k_c / P}{k_c (C_a - 1)} \quad , \qquad \mathcal{I} = \frac{2}{P + A} (\mathcal{I}_R + \mathcal{I}_C)$$

where b identifies a group and C_i or C_a its size on the corresponding side of the bipartite network. Setting all nodes to one group gives global nestedness $\mathcal{N}$, measuring how well one ladder describes the entire matrix; the same local–global distinction appears in economic networks [17]. Significance was assessed with 100 Bernoulli matrices satisfying $\Pr(M_{pa}^{\text{null}} = 1) = \min(k_p k_a / E, 1)$, $E = \sum_{pa} M_{pa}$, which preserve degree heterogeneity in expectation. Empty nodes were removed and every null matrix was re-clustered, allowing group structure to arise by chance.

Fitness and complexity: what a species has accumulated, what a drug requires

The Fitness and Complexity framework captures the idea that not all comparative resistance links are equally informative [4]. A species that defeats many commonly overcome antibiotics may be less remarkable than one that defeats fewer drugs, including antibiotics that few other species can defeat. The *fitness* F_p of a specie p therefore reflects both the breadth of its comparative resistance profile and the difficulty of the antibiotics it defeats. Conversely, the *complexity* Q_a of an antibiotic a depends not simply on how rarely a drug is defeated, but on which species defeat it. A drug has high complexity only when it is defeated exclusively by species with broad comparative resistance profiles; even one species with a narrow profile substantially lowers its complexity. Fitness and complexity are therefore estimated jointly, with each helping to determine the other. Starting from $F_p^{(0)} = Q_a^{(0)} = 1$, the pipeline iterates

$$\widetilde{F}_p^{(n)} = \sum_a M_{pa} Q_a^{(n-1)} \quad , \qquad \widetilde{Q}_a^{(n)} = \left(\sum_p \frac{M_{pa}}{F_p^{(n-1)}} \right)^{-1}$$

normalizing F and Q by their means at each step. Because heterogeneous non-nested structures can produce collapsed rankings, scores were computed separately within each group and standardized before pooling. They therefore describe position on a local ladder, not an absolute scale shared across groups. Fitness is a network score, unrelated to evolutionary fitness; complexity is not drug potency or clinical efficacy.

Two external classifications were used only after scoring: the WHO Access, Watch and Reserve stewardship tiers [5] for antibiotic complexity, and the ESKAPEE group, which extends the ESKAPE mnemonic to include *Escherichia coli* [6], for pathogen fitness. Neither entered the network, grouping or scores.

The antibiotic similarity network and prospective forecasting

Antibiotics repeatedly linked to the same species have similar comparative resistance profiles. Such co occurrence may reflect shared mechanisms, linked determinants, correlated selection, or lineage structure, but it does not identify the cause [2, 18]. Projecting the bipartite network onto its antibiotic side makes the empirical similarity explicit: $W^A_{ab} = \sum_p M^{(t)}_{pa} M^{(t)}_{pb}$, giving a weighted similarity network over the 40 antibiotics. A species that already defeats many of a drug's neighbors has high network exposure to that drug. This quantity is an empirical proximity to the next rung, and not meant as a direct evidence of a shared mechanism. For each origin year $t = 2019, \dots, 2023$, all quantities were rebuilt using only data from 2004 through t. For every sufficiently observed pair (p, a) with no link at t, exposure was the weighted share of drug a's neighbourhood already defeated by species p:

$$\rho^A_{pa} = \frac{\sum_o W^A_{ao} M^{(t)}_{po}}{\sum_o W^A_{ao}}$$

The outcome was whether the same sufficiently tested pair had $\mathrm{RCA}_{pa} \geq 1$ in year $t + 1$. The score contains no fitted coefficient, training step or tuned threshold. Discrimination was measured by AUC against two baselines: the drug's current fraction of linked species, and the candidate pair's current RCA_{pa} (its proximity to the threshold). Significance used 200 node-label permutations. To test persistence, the 2019 network, scores and candidate set were frozen and evaluated unchanged against 2020–2024, restricting each evaluation to pairs sufficiently tested in that target year.

Results

Comparative-resistance profiles are nested: each group is a ladder

Before interpreting the bipartite network as a resistance ladder and applying Fitness and Complexity, we first tested whether comparative resistance profiles displayed the expected nested structure. Pooling 2015–2024 yielded 1,932 sufficiently observed cells in a 139-by-40 network, of which 991 carried a comparative-resistance link. Within the five groups, nestedness was pronounced: $\mathcal{I}^* = 0.3855$ against 0.1453 ± 0.0229 for degree-matched randomization ($z = 10.51$). Narrow comparative-resistance profiles sat inside broader ones, and drugs at the top of each group were linked only to species that defeated almost everything else in it. The groups therefore formed local resistance ladders consistent with partially ordered accumulation [8, 9]. Applied to M as a whole, the same statistic was low and below the null ($\mathcal{N} = 0.0496$ against 0.0844 ± 0.0081, $z = -4.30$); $\mathcal{I}^*/\mathcal{N} = 7.77$. This is consistent with combining organisms that differ in intrinsic barriers, resistance repertoires and testing spectra [13, 14]: a *Klebsiella* and an *Enterococcus* do not lie on one biologically interpretable axis. As a qualitative check, we found that the five data-derived groups matched recognizable organism–drug domains: one large Gram-negative group; separate staphylococcal and enterococcal groups; and two anaerobe groups centred on *Clostridia* and *Bacteroides*, each paired with the drugs represented in its testing profile.

Fitness and complexity place species and drugs on the ladder

Having established that comparative resistance profiles form local ladders, we applied Fitness and Complexity separately within each group. Within each group the equations converged to a stable fitness for every species and complexity for every drug. We then asked whether the resulting rankings aligned with biological and clinical classifications that had been withheld from the construction. Antibiotic complexity increased across WHO AWaRe tiers (Spearman $\rho = 0.5895$, $p = 9.84 \times 10^{-5}$, 38 drugs; $\rho = 0.6244$, $p = 8.49 \times 10^{-4}$ in the largest group), aligning the network ranking with the stewardship gradient [5] (Figure 1a). The species ranking showed a similarly coherent pattern. ESKAPEE taxa [6] occupied the upper rungs: median standardised fitness was 1.044 for 16 matching taxa and -0.387 for 107 others (one-sided Mann–Whitney $p = 4.60 \times 10^{-6}$; Figure 1b). These associations provide external consistency, not clinical validation: both scores use only the topology of comparative-resistance links.

Comparative-resistance profiles expand toward similar drugs, and the next rung can be anticipated

The bipartite network provides more than a static ranking of species and antibiotics. Its structure can also be used to study how comparative resistance profiles expand. Antibiotics linked to many of the same species have similar positions in the network and may therefore represent nearby opportunities for future resistance. We made this relationship explicit by projecting the bipartite graph onto its antibiotic side, connecting drugs according to the number of species linked to both. (Figure 2a). The evolution of *Klebsiella pneumoniae* illustrates this local expansion. Comparative-resistance links expanded locally from 12 drugs in 2014 to 17 in 2019 and 21 in 2024, while distant regions remained unlinked (Figure 2b–d). This pattern shows the resistance ladder in motion and motivates the use of network proximity to anticipate the next rung. It remains a descriptive pattern and does not establish molecular acquisition or transmission.

We therefore defined network exposure as the extent to which a species was already linked to antibiotics similar to a candidate drug. The rolling origin evaluation contained 1,839 eligible pathogen–drug pair-years, of which 131 (7.12%) crossed the comparative-resistance threshold in the following year. The exposure ranking clearly separated absolute event rates: 51 of 307 pairs (16.61%) in the top exposure sixth crossed the threshold within one year, compared with 14 of 307 (4.56%) in the bottom sixth, a 3.6-fold difference (Figure 3a). Network exposure, rebuilt at each origin from raw co-occurrence weights in M, reached a pooled AUC of 0.7076 (origin-specific 0.6688–0.7703), compared with 0.5123 for drug prevalence (Figure 3b). An AUC of 0.7076 means that a future event outranked a non-event 70.76% of the time; it measures ranking performance, not individual probability. Exposure also exceeded all 200 node-label permutations ($p < 0.005$). Per-drug AUC was highest for ceftazidime, gentamicin, meropenem, cefepime and ceftriaxone, reaching 0.89 for ceftazidime (Figure 3c). We call these drugs "contagion hubs": a descriptive network label for neighborhoods that best announce the next comparative-resistance event. Each estimate rests on 5–9 events and remains exploratory.

It should be noted that the signal persisted beyond one year. With network, scores and candidate set frozen at 2019, AUC remained 0.681–0.742 across one- to five-year horizons, while the prevalence baseline stayed near chance (0.41–0.58). Results were also robust to the isolate threshold: repeating the pipeline at $n_{\min} = 10, 30, 50$ preserved in-group nestedness and ESKAPEE separation. The default $n_{\min} = 30$ gave the strongest AWaRe association ($\rho = 0.590$, versus 0.547 and 0.442) and highest back-test AUC (0.708, versus 0.638 and 0.702).

Impact of the work

What this adds to surveillance practice. A surveillance report answers "how much resistance is there?", not "where is it going?". From the same counts, this framework adds three views: a *position* for every species and drug on its local ladder; an antibiotic *similarity network* built from comparative-resistance behaviour rather than chemical class; and a *ranked watch-list* of pairs that lie near existing comparative-resistance links. Contagion hubs also nominate drugs for prospective evaluation as early-warning sentinels when testing capacity is limited.

Interpretability is a design requirement, not a claim. The chain is short and inspectable (counts $\rightarrow r_{pa} \rightarrow \mathrm{RCA}_{pa} \rightarrow M \rightarrow$ similarity network $\rightarrow$ exposure score): no classifier is trained, coefficient fitted or threshold tuned. A pair ranks highly because specific drugs in its neighbourhood are already linked to that species, and this rationale can be checked against the isolate counts and challenged by a microbiologist. The analysis can be recomputed as soon as new counts arrive.

The Resistome Atlas. The interactive dashboard (`https://giordano-demarzo.github.io/resistome-atlas/`) is delivered with a deterministic end-to-end script and output tables. Users can follow a pathogen's comparative-resistance profile through the antibiotic similarity network, inspect local ladder positions alongside AWaRe labels, and view the drugs it is closest to defeating. Surveillance teams can identify climbing organisms, stewardship groups can inspect restricted drugs near existing links, and laboratories can prioritise candidate pairs for confirmatory testing. Dashboard probabilities come from a separate one-variable calibration of exposure, not from the parameter-free evaluation reported here; no output is clinical guidance.

Deployment beyond this dataset. The pipeline needs only isolate counts by species, drug and year. Because comparative resistance is defined relative to the analysed population, fractions, margins, groups and networks must be recomputed for each country, region or time window; one hospital network need not share the global ladder. Temporal re-estimation is demonstrated here, whereas geographical re-estimation remains a deployment opportunity because the present table pools countries. Genomic integration could test ladder-implied relationships against known resistance genes and mutations, while recognising that genotype–phenotype correspondence is incomplete [19].

Limits. Comparative resistance is relative: a pair may be flagged at modest prevalence, or remain unremarkable despite clinically severe absolute resistance. ATLAS is a surveillance sample rather than a population-random sample, so representativeness may be limited, as in laboratory-based surveillance more generally [20]. A zero in M also merges observed pairs below the threshold with insufficiently tested pairs, so the network partly reflects testing coverage. The number of groups is fixed at five, the null preserves degree only in expectation, and AWaRe and ESKAPEE provide external consistency rather than clinical validation. AUC measures ranking, not individual risk; changes in comparative status may reflect shifting margins or sampling composition as well as biology; and nothing here establishes causation, transmission or treatment failure.

Tables/figures

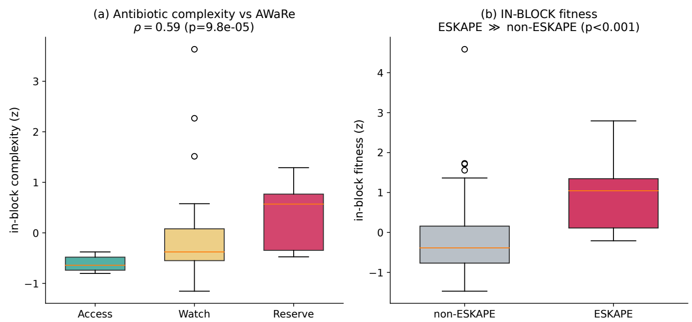

Figure 1: Network rankings recover known clinical and biological groups. Species fitness increases when a species is comparatively resistant to many antibiotics, especially to drugs resisted by few other species; antibiotic complexity is higher when a drug is comparatively resisted only by high-fitness species. Both scores are calculated from the network alone, without using external labels. (a) Antibiotic complexity increases from the WHO Access to Watch and Reserve groups. (b) ESKAPEE species have higher fitness and occupy the upper rungs of their local resistance ladders. Fitness is a network score, not evolutionary fitness, and complexity does not measure drug potency.

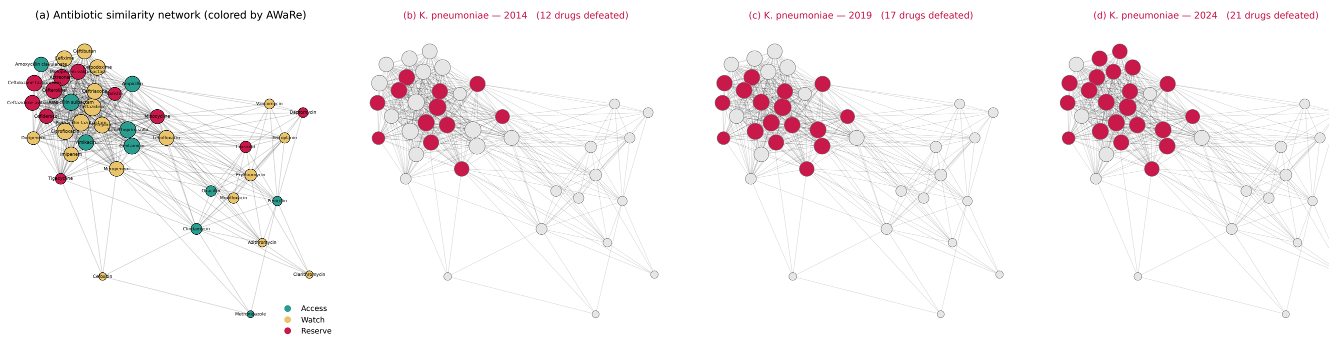

Figure 2: Antibiotic similarity network and expansion of resistance in *Klebsiella pneumoniae*. (a) Antibiotics (nodes) are connected when they are comparatively resisted by the same species: grey lines (links) show these similarities, while node colours indicate the WHO Access, Watch and Reserve classes for illustration only. (b–d) The same network is shown for *K. pneumoniae* in 2014, 2019 and 2024, highlighting the antibiotics to which it has a comparative-resistance link. Over time, new links tend to appear near antibiotics that were already linked, suggesting that comparative resistance expands locally across the antibiotic similarity network.

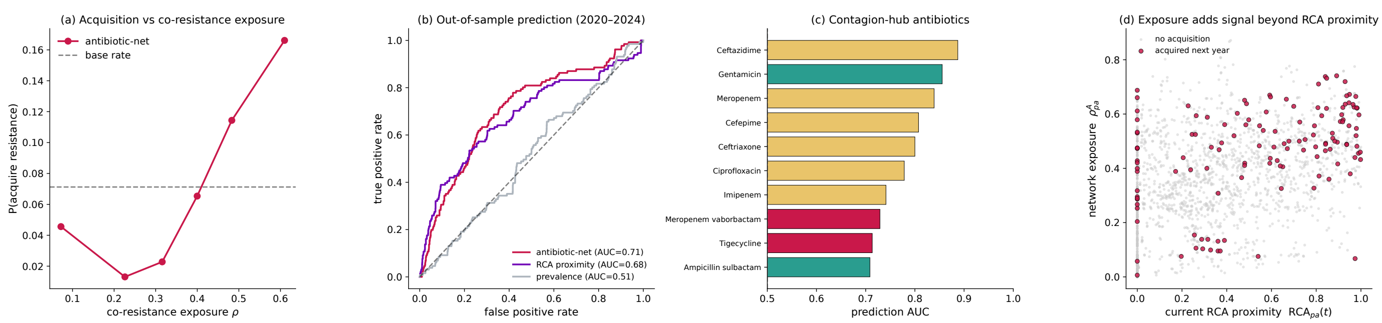

Figure 3: Forecasting the next drug. Network exposure anticipates new comparative-resistance links. For each pathogen–antibiotic pair that is not yet linked in year t, exposure measures how strongly that pathogen is already linked to antibiotics similar to the candidate drug. (a) The plot shows the observed fraction of pairs that develop a new comparative-resistance link in the following year: this fraction (y-axis) is higher when exposure is high (x-axis). (b) Exposure ranks future links better than the pair's current RCA and than how commonly the drug is already linked across pathogens (AUC 0.71 vs 0.68 and 0.51). (c) Drug-specific AUCs identify the antibiotics whose neighbourhoods in the antibiotic similarity network are most informative for predicting which pathogens will develop a new comparative-resistance link to that drug in the following year.

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